CHAPTER 8

TRIANGULATION

Introduction

We work in this chapter with a fixed o-minimal expansion $(R, <, \mathcal{S})$ *of an ordered real closed field* $(R, <, 0, 1, +, -, \cdot)$.

We point out, however, that most of Section 1 is of a purely semilinear character, and makes sense over any ordered field. In Section 2 we prove the main result in this chapter, the triangulation theorem, which allows us to reduce many questions to the semilinear case. In Section 3 we use triangulation in this way to show that a definable continuous function $f: A \to R$ on a definable closed subset A of a definable set B can be continuously and definably extended to B. This result is needed in the next two chapters on trivialization and quotient spaces.

§1. Simplexes and complexes

For simplicity, we denote the ordered field $(R, <, 0, 1, +, -, \cdot)$ just by R. In this section the role of R is mainly that of an ordered vector space over itself. In particular we will only deal with subsets of R^n that are semilinear over R, in the sense of Chapter 1, (7.9). We call a map $f: A \to R^n$ with $A \subseteq R^m$ **semilinear** if its graph $\Gamma(f) \subseteq R^{m+n}$ is semilinear.

(1.1) An **affine** subspace of R^n of dimension d is by definition a translate $L + a$ of a linear subspace L of R^n of dimension d. For $d = 1$ this is also called a **line**, and for $d = 2$ we speak of a **plane**. Given points $a_0, a_1, \ldots, a_k$ in R^n we have

$$\begin{aligned} a_0 + R(a_1 - a_0) + \cdots + R(a_k - a_0) &= \left\{\sum t_i a_i :\ t_i \in R,\ \sum t_i = 1\right\} \\ &= \text{the smallest affine subspace containing } a_0, \ldots, a_k. \end{aligned}$$

This affine subspace is called the **affine span** of $a_0, \ldots, a_k$.

(1.2) A tuple $a_0, \ldots, a_k$ of points in R^n is called **affine independent** if the affine span of $a_0, \ldots, a_k$ has dimension k. For $k = 0$ this is always the case, for $k = 1$ it means that $a_0 \neq a_1$, for $k = 2$ that a_0, a_1, a_2 are distinct and not on a line.

Affine independence of $a_0, a_1, \ldots, a_k$ is equivalent to the condition that the k vectors $a_1 - a_0, \ldots, a_k - a_0$ are linearly independent.

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Let $a_0, \ldots, a_k$ be affine independent, $a_i \in R^n$. Then the map $(t_0, \ldots, t_k) \mapsto \sum t_i a_i$ is a homeomorphism from the affine subspace $\{(t_0, \ldots, t_k) \in R^{k+1} : \sum t_i = 1\}$ of R^{k+1} onto the affine span of $a_0, \ldots, a_k$. In particular, each point x in the affine span of $a_0, \ldots, a_k$ has a *unique* representation as

$$x = t_0 a_0 + \cdots + t_k a_k \quad \text{with } \sum t_i = 1,$$

and $t_0, \ldots, t_k$ are called the **affine coordinates** of the point x with respect to $a_0, \ldots, a_k$. (Sometimes also called **barycentric** coordinates.)

(1.3) A set $S \subseteq R^n$ is called **convex** if for every two distinct points a, b in S the line segment $[a, b] := \{ta + (1-t)b : 0 \leq t \leq 1\}$ is contained in S. The **convex hull** of a set $S \subseteq R^n$ is the smallest convex subset of R^n that contains S, and is easily seen to equal the set of all points $\sum t_i a_i$ with $a_0, \ldots, a_k \in S$ and $\sum t_i = 1$, $t_i \geq 0$.

Given a convex set C in R^n, a point $p \in C$ is called an **extreme** point of C if for all $p_1, p_2 \in C$ such that $p = (p_1 + p_2)/2$ we have $p = p_1 = p_2$.

(1.4) Definitions. An affine independent tuple of points $a_0, a_1, \ldots, a_k \in R^n$ is said to **span the simplex**

$$(a_0, \ldots, a_k) := \left\{\sum t_i a_i : \text{ all } t_i > 0, \sum t_i = 1\right\} \subseteq R^n.$$

We call $(a_0, \ldots, a_k)$ a **k-simplex** in R^n. Note that it is *open* in the affine span of $a_0, \ldots, a_k$. Its dimension as a semilinear set is easily seen to be k.

The (topological) closure of $(a_0, \ldots, a_k)$ in R^n is denoted by $[a_0, \ldots, a_k]$, so that

$$\begin{aligned} [a_0, \ldots, a_k] &:= \left\{\sum t_i a_i : \text{ all } t_i \geq 0, \sum t_i = 1\right\} \\ &= \text{the convex hull of } \{a_0, \ldots, a_k\}. \end{aligned}$$

One easily checks that $a_0, \ldots, a_k$ are exactly the extreme points of $[a_0, \ldots, a_k]$. We call $a_0, \ldots, a_k$ the **vertices** of $(a_0, \ldots, a_k)$ and also of $[a_0, \ldots, a_k]$. Instead of "S is the set of vertices of the simplex σ" we also say "S spans σ".

A **face** of $(a_0, \ldots, a_k)$ is a simplex spanned by a nonempty subset of $\{a_0, \ldots, a_k\}$. Note that distinct nonempty subsets of $\{a_0, \ldots, a_k\}$ span *disjoint* faces, and that $[a_0, \ldots, a_k]$ is the (disjoint) union of the faces of $(a_0, \ldots, a_k)$.

Given simplexes σ and τ we write $\sigma < \tau$ if σ is a **proper** face of τ, that is, σ is a face of τ and $\sigma \neq \tau$.

The **barycenter** $b(\sigma)$ of a k-simplex $\sigma = (a_0, \ldots, a_k)$ is the point

$$b(\sigma) = \frac{1}{k+1}(a_0 + \cdots + a_k),$$

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which belongs to σ.

Examples of closures of simplexes.

$$
\begin{aligned}
[a_0] &= (a_0) = \{a_0\}; \text{ however, } a_i \notin (a_0, \ldots, a_k) \text{ if } k > 0; \\
[a_0, a_1] &= (a_0, a_1) \cup \{a_0\} \cup \{a_1\}, \text{ the line segment between} \\
&\quad \text{distinct points } a_0 \text{ and } a_1; \\
[a_0, a_1, a_2] &= \text{the "triangle" spanned by distinct points } a_0, a_1, a_2 \text{ not on a line;} \\
[a_0, a_1, a_2, a_3] &= \text{the "tetrahedron" spanned by four distinct points } a_0, a_1, a_2, a_3 \\
&\quad \text{that do not all lie on one plane.}
\end{aligned}
$$

(1.5) Definitions. A **complex in** R^n is a finite collection K of simplexes in R^n, such that for all $\sigma_1, \sigma_2 \in K$,

$$
\begin{aligned}
\text{either } \mathrm{cl}(\sigma_1) \cap \mathrm{cl}(\sigma_2) &= \emptyset, \\
\text{or } \mathrm{cl}(\sigma_1) \cap \mathrm{cl}(\sigma_2) &= \mathrm{cl}(\tau)
\end{aligned}
$$

for some common face τ of σ_1 and σ_2. (This τ is not required to belong to K, which makes our definition somewhat more general than is usual. The usual definition of "complex" corresponds to what we call "closed complex" below.)

Note that distinct simplexes in K are disjoint. We put

$$
\begin{aligned}
|K| &:= \text{ union of the simplexes of } K \text{, a bounded semilinear subset of } R^n; \\
\mathrm{Vert}(K) &:= \text{ the set of vertices of the simplexes in } K \text{, a finite subset of } R^n.
\end{aligned}
$$

We also call $|K|$ the **polyhedron** spanned by the complex K. Note: the notational conflict that $|K|$ also denotes the number of simplexes of K is always resolved in favor of "$|K| =$ polyhedron spanned by K". (All bounded semilinear sets in R^n are polyhedrons, that is, of the form $|K|$ for a suitable complex K in R^n, see (2.14), exercise 2.)

A subset of a complex K in R^n is also a complex in R^n, and is called a **subcomplex of** K.

A complex K is called **closed** if it contains with each simplex all its faces. Equivalently, a closed complex in R^n is a finite collection K of disjoint simplexes in R^n such that each face of a simplex in K also belongs to K. Note also that a complex K in R^n is closed iff $|K|$ is closed in R^n.

Given a complex K in R^n we put $\overline{K} :=$ the set of faces of the simplexes in K. So $\overline{K}$ is clearly a closed complex in R^n, with K as a subcomplex, and

$$
\mathrm{cl}\,|K| = |\overline{K}|.
$$

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Let L be a subcomplex of K. Then $|L|$ is closed in $|K|$ if and only if $\overline{L} \cap K = L$. In that case we also say that L is **closed in** K.

Note that a complex K in R^n is completely determined by the finite set $\mathrm{Vert}(K) \subseteq R^n$ and by the collection of subsets of $\mathrm{Vert}(K)$ that span simplexes in K.

(1.6) Let K and L be complexes in R^m and R^n respectively.

Then a **vertex map** $V: K \to L$ is a map $V: \mathrm{Vert}(K) \to \mathrm{Vert}(L)$ such that whenever $(a_0, \ldots, a_k)$ is a k-simplex of K, then $\{V(a_0), \ldots, V(a_k)\}$ spans a simplex of L. (We allow $V(a_i) = V(a_j)$ for $i \neq j$.)

Such a vertex map $V: K \to L$ determines a continuous map $|V|: |K| \to |L|$ by:

$$|V|\left(\sum t_i a_i\right) = \sum t_i V(a_i)$$

where $(a_0, \ldots, a_k)$ is a k-simplex of K, $t_i > 0$, $\sum t_i = 1$.

To show that $|V|$ is continuous, show first that the restriction of $|V|$ to the closure of $(a_0, \ldots, a_k)$ in $|K|$ is continuous. Then use the fact that if $f: X \to Y$ is a map between topological spaces X and Y, and $X = X_1 \cup \cdots \cup X_k$ a covering of X by finitely many closed sets X_i such that each restriction $f|X_i: X_i \to Y_i$ is continuous, then f is continuous. This general fact will be used frequently.

Note also that the map $|V|$ is semilinear, and that if $W: L \to M$ is a second vertex map, then the composition $W \circ V: K \to M$ is a vertex map such that

$$|W \circ V| = |W| \circ |V|.$$

We now state our main result concerning triangulation.

(1.7) TRIANGULATION THEOREM. *Each definable set $S \subseteq R^m$ is definably homeomorphic to a polyhedron $|K|$ for some complex K in R^m.*

This theorem expresses the truly remarkable fact that the (definable) topology of a definable set can be completely described in *finite combinatorial terms*. To see this, let K be a complex, and consider its *scheme*

$$\left(\mathrm{Vert}(K), \{S \subseteq \mathrm{Vert}(K): \ S \text{ spans a simplex of } K\}\right),$$

which is just a finite set equipped with a collection of subsets, a finite combinatorial object *par excellence*. For L to be a complex with isomorphic scheme means there is a bijective vertex map $V: K \to L$ whose inverse is also a vertex map. Such a map V induces a semilinear homeomorphism $|V|: |K| \to |L|$. In this sense, the scheme of K, as a finite combinatorial object, determines the topology of the polyhedron $|K|$ up to semilinear homeomorphism.

We prove this theorem in a more precise version in the next section. In the rest of this section we consider some elementary operations on complexes.

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(1.8) BARYCENTRIC SUBDIVISION. Let K be a complex. A **K-flag** is a sequence

$$\mathfrak{F}:\ \sigma_0 < \sigma_1 < \cdots < \sigma_f \quad (f \geq 0)$$

of simplexes, with $\sigma_f \in K$ and each σ_i a proper face of the σ_j with $j > i$.

We do not require $\sigma_i \in K$ for $i < f$, or that σ_i is of dimension i.

To such a K-flag $\mathfrak{F}$ we associate an f-simplex $b(\mathfrak{F}) := \big(b(\sigma_0), \ldots, b(\sigma_f)\big)$ whose vertices are the barycenters of the simplexes of $\mathfrak{F}$. One easily checks that $b(\mathfrak{F})$ is indeed an f-simplex, that $b(\mathfrak{F}) \subseteq \sigma_f$, and that if $\mathfrak{F}_1$ and $\mathfrak{F}_2$ are distinct K-flags then $b(\mathfrak{F}_1)$ and $b(\mathfrak{F}_2)$ are disjoint.

The (first) **barycentric subdivision of** K is the complex K' whose simplexes are the simplexes $b(\mathfrak{F})$ associated to the K-flags $\mathfrak{F}$. One checks easily that the $b(\mathfrak{F})$ indeed form a complex, and that $|K'| = |K|$.

(1.9) We need one further general construction.

Let $(a_0, \ldots, a_k)$ be a k-simplex in R^n, and $r_i, s_i \in R$, $r_i \leq s_i$ for $i = 0, \ldots, k$, and $r_j < s_j$ for some j. Put $b_i := (a_i, r_i)$, $c_i := (a_i, s_i) \in R^{n+1}$. Then $(b_0, \ldots, b_k)$, $(c_0, \ldots, c_k)$ are k-simplexes in R^{n+1}. One also easily checks that if $b_i \neq c_i$ (that is, $r_i < s_i$) then $(b_0, \ldots, b_i, c_i, \ldots, c_k)$ is a $(k+1)$-simplex. We now construct a complex L in R^{n+1} such that $|L|$ is the set of points "between $[b_0, \ldots, b_k]$ and $[c_0, \ldots, c_k]$".

(1.10) LEMMA. *Let L consist of all $(k+1)$-simplexes $(b_0, \ldots, b_i, c_i, \ldots, c_k)$ with $b_i \neq c_i$, and all faces of these $(k+1)$-simplexes. Then L is a closed complex and*

$$\begin{aligned} |L| &= \Big\{ t\big(t_0 b_0 + \cdots + t_k b_k\big) + (1-t)\big(t_0 c_0 + \cdots + t_k c_k\big) : \\ &\qquad 0 \leq t \leq 1,\ t_i \geq 0,\ \sum t_i = 1 \Big\} \\ &= \textit{convex hull of } \big\{ b_0, \ldots, b_k, c_0, \ldots, c_k \big\}. \end{aligned}$$

PROOF. The proof of this lemma is rather long, and is divided into four parts.

PART I. In this part we show L *is a closed complex*. Let σ and τ be faces of $(b_0, \ldots, b_i, c_i, \ldots, c_k)$ and $(b_0, \ldots, b_j, c_j, \ldots, c_k)$ respectively, $r_i < s_i$, $r_j < s_j$.

Suppose $\sigma \cap \tau \neq \emptyset$. We have to show that $\sigma = \tau$. This is clear if $i = j$ since then σ and τ are faces of the same simplex. So assume $i < j$, and let $x \in \sigma \cap \tau$. Write

$$\begin{aligned} x &= t_0 b_0 + \cdots + t_i b_i + t c_i + t_{i+1} c_{i+1} + \cdots + t_k c_k, \text{ with } t + t_0 + \cdots + t_k = 1, \\ x &= u_0 b_0 + \cdots + u_{j-1} b_{j-1} + u b_j + u_j c_j + \cdots + u_k c_k, \text{ with } u + u_0 + \cdots + u_k = 1. \end{aligned}$$

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Applying the projection map $\pi: R^{n+1} \to R^n$ to these expressions for x and considering the affine coordinates of $\pi(x)$ with respect to $a_0, \ldots, a_k$ we get

$$(*) \quad t_i + t = u_i,\ t_j = u_j + u, \text{ and } t_\lambda = u_\lambda \text{ for } \lambda \in \{0, \ldots, k\},\ \lambda \neq i,\ \lambda \neq j.$$

Also the $(n+1)^{\text{th}}$ coordinate of x satisfies

$$\begin{aligned} x_{n+1} &= t_0 r_0 + \cdots + t_i r_i + t s_i + t_{i+1} s_{i+1} + \cdots + t_k s_k \\ &= u_0 r_0 + \cdots + u_{j-1} r_{j-1} + u r_j + u_j s_j + \cdots + u_k s_k. \end{aligned}$$

Subtracting $t_0 r_0 + \cdots + t_{i-1} r_{i-1} + t_{j+1} s_{j+1} + \cdots + t_k s_k$ from both sums yields

$$t_i r_i + t s_i + \sum_{i+1 \leq \lambda \leq j} t_\lambda s_\lambda = \sum_{i \leq \lambda \leq j-1} u_\lambda r_\lambda + u r_j + u_j s_j.$$

Subtracting the right side from the left and using $(*)$ we get

$$t(s_i - r_i) + \sum_{i+1 \leq \lambda \leq j-1} t_\lambda (s_\lambda - r_\lambda) + u(s_j - r_j) = 0,$$

in which each term on the left is nonnegative. Hence each term equals zero, so $t = u = 0$, and $t_\lambda = 0$ whenever $i+1 \leq \lambda \leq j-1$ and $r_\lambda < s_\lambda$. The vertices of σ are exactly the points among $b_0, \ldots, b_i, c_i, \ldots, c_k$ with respect to which x has a nonzero affine coordinate, and similarly with τ and $b_0, \ldots, b_j, c_j, \ldots, c_k$. This shows that σ and τ have the same vertices, so that $\sigma = \tau$.

PART II. In this part we prove that $|L|$ contains the set

$$\left\{ t\bigl(t_0 b_0 + \cdots + t_k b_k\bigr) + (1-t)\bigl(t_0 c_0 + \cdots + t_k c_k\bigr) :\ 0 \leq t \leq 1,\ t_i \geq 0,\ \sum t_i = 1 \right\}.$$

Let $x = t\bigl(t_0 b_0 + \cdots + t_k b_k\bigr) + (1-t)\bigl(t_0 c_0 + \cdots + t_k c_k\bigr)$, $0 \leq t \leq 1, t_i \geq 0$, $\sum t_i = 1$. We have to show that $x \in |L|$. Clearly the $(n+1)^{\text{th}}$ coordinate x_{n+1} of x satisfies

$$t_0 r_0 + \cdots + t_k r_k \leq x_{n+1} \leq t_0 s_0 + \cdots + t_k s_k,$$

so there is an $i \in \{0, \ldots, k\}$ such that $r_i < s_i$ and

$$\begin{aligned} &t_0 r_0 + \cdots + t_{i-1} r_{i-1} + t_i r_i + t_{i+1} s_{i+1} + \cdots + t_k s_k \leq x_{n+1} \\ &\leq t_0 r_0 + \cdots + t_{i-1} r_{i-1} + t_i s_i + t_{i+1} s_{i+1} + \cdots + t_k s_k. \end{aligned}$$

Hence $x_{n+1} = t_0 r_0 + \cdots + t_{i-1} r_{i-1} + a + t_{i+1} s_{i+1} + \cdots + t_k s_k$, with $t_i r_i \leq a \leq t_i s_i$. Then we can write $a = u r_i + v s_i$ with $u, v \geq 0$ and $u + v = t_i$. Thus

$$x_{n+1} = t_0 r_0 + \cdots + t_{i-1} r_{i-1} + u r_i + v s_i + t_{i+1} s_{i+1} + \cdots + t_k s_k.$$

Also

$$\pi(x) = \sum t_i a_i = t_0 a_0 + \cdots + t_{i-1} a_{i-1} + u a_i + v a_i + t_{i+1} a_{i+1} + \cdots + t_k a_k.$$

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Therefore

$$x = t_0 b_0 + \cdots + t_{i-1} b_{i-1} + u b_i + v c_i + t_{i+1} c_{i+1} + \cdots + t_k c_k,$$

since the right hand side has the same $(n+1)^{\text{th}}$ coordinate and the same projection in R^n as x. This expression for x shows that $x \in [b_0, \dots, b_i, c_i, \dots, c_k]$, and finishes the proof of part II.

Part III. The set $S := \left\{ t \sum t_i b_i + (1-t) \sum t_i c_i : \ 0 \le t \le 1, \ t_i \ge 0, \ \sum t_i = 1 \right\}$ is convex.

To prove this, first a preliminary remark.

For a point $x \in S$, $x = t \sum t_i b_i + (1-t) \sum t_i c_i$ with $0 \le t \le 1$, $t_i \ge 0$, $\sum t_i = 1$, we have $\pi(x) = \sum t_i a_i \in [a_0, \dots, a_k]$. Hence, given a point $p = \sum t_i a_i \in [a_0, \dots, a_k]$, $t_i \ge 0$, $\sum t_i = 1$, the vertical line $\pi^{-1}\{p\} = (p, 0) + R e_{n+1}$ intersects S exactly in the line segment $\left[\sum t_i b_i, \sum t_i c_i \right]$, and the set of $(n+1)^{\text{th}}$ coordinates of the points on this line segment equals $\left[\sum t_i r_i, \sum t_i s_i \right]$.

Let now $x, y \in S$, and consider a convex combination $z = ux + vy$, $u, v \ge 0$, $u + v = 1$. We have to show that $z \in S$. Write $x = t \sum t_i b_i + (1-t) \sum t_i c_i$ as above, and similarly $y = t' \sum t'_i b_i + (1-t') \sum t'_i c_i$. Then

$$\pi(z) = u\pi(x) + v\pi(y) = \sum \left(u t_i + v t'_i \right) a_i \in [a_0, \dots, a_k],$$

and the $(n+1)^{\text{th}}$ coordinates satisfy

$$z_{n+1} = u x_{n+1} + v y_{n+1}, \ \sum t_i r_i \le x_{n+1} \le \sum t_i s_i$$

and $\sum t'_i r_i \le y_{n+1} \le \sum t'_i s_i$. Hence $\sum (u t_i + v t'_i) r_i \le z_{n+1} \le \sum (u t_i + v t'_i) s_i$, so that by the preliminary remark we have $z \in S$.

Part IV. Rest of the proof. The set S from part III obviously contains the points $b_0, \dots, b_k, c_0, \dots, c_k$, hence it contains the convex hull of these points. Finally, the convex hull of these points contains each set $[b_0, \dots, b_i, c_i, \dots, c_k]$ with $b_i \ne c_i$, hence it contains the union $|L|$ of these sets. This completes the circle of inclusions, and thereby the proof of the lemma. □

(1.11) Remarks. One should note that the complex L of our lemma depends on the listing of the vertices of $(a_0, \dots, a_k)$ in the order indicated: first a_0, then a_1, etc. On the other hand, its polyhedron $|L|$ is independent of this ordering. Note also that the simplexes $(b_0, \dots, b_k)$ and $(c_0, \dots, c_k)$ are in L, and that $\pi(\sigma)$ is a face of $(a_0, \dots, a_k)$ for each $\sigma \in L$.

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(1.12) Exercises.

To state the first exercise, define a map $f: R^m \to R^n$ to be **affine** if it is of the form $f(x) = L(x) + a$ where $L: R^m \to R^n$ is linear, and $a \in R^n$. Note that for $n = 1$ this is just what we called an affine function in Chapter 1, Section 7.

1. Show that the composition of affine maps is affine, and that if $f: R^m \to R^n$ is affine, then $f\big(\sum t_i a_i\big) = \sum t_i f(a_i)$ for $a_i \in R^m$, and $t_i \in R$ with $\sum t_i = 1$. Conclude that if two affine maps $R^m \to R^n$ take the same value at points $a_0, \ldots, a_k \in R^m$, they agree on the affine subspace of R^m spanned by $a_0, \ldots, a_k$.

2. Let $T : R^m \to R^m$ be an R-linear automorphism, or a translation, and let $a_0, \ldots, a_k$ be an affine independent tuple of points in R^m. Show that the tuple $T(a_0), \ldots, T(a_k)$ is affine independent and that

$$T\big((a_0, \ldots, a_k)\big) = \big(T(a_0), \ldots, T(a_k)\big).$$

3. Let σ be a k-simplex in R^m. Show that there are $m-k$ polynomials $f_1, \ldots, f_{m-k} \in R[X_1, \ldots, X_m]$ and $k+1$ polynomials $g_0, \ldots, g_k \in R[X_1, \ldots, X_m]$, with all f's and g's of degree 1, such that

$$\sigma = \big\{x \in R^m : \ f_i(x) = 0, \ g_j(x) > 0, \ 1 \leq i \leq m-k, \ 0 \leq j \leq k\big\},$$

and

$$\mathrm{cl}(\sigma) = \big\{x \in R^m : \ f_i(x) = 0, \ g_j(x) \geq 0, \ 1 \leq i \leq m-k, \ 0 \leq j \leq k\big\}.$$

Hint: let the affine space spanned by the vertices of σ be given by the system of equations $f_i(x) = 0$ $(1 \leq i \leq m-k)$.

4. Let $V : K \to L$ be a vertex map between complexes K and L, and let $\sigma \in K$. Show that $|V|(\sigma)$ is the simplex of L spanned by $\big\{V(p) : \ p \text{ is a vertex of } \sigma\big\}$.

5. Let K be a complex in R^m, $\sigma \in K$ and $\dim(\sigma) = \dim\big(|K|\big)$. Show that then σ is open in $|K|$.

6. (Carathéodory) Show that the convex hull of a set $S \subseteq R^m$ is the union of the convex hulls of its finite subsets of size $\leq m+1$. (Hint: assume $|S| = m+2$.)

7. Show that the convex hull of a definable set in R^m is definable.

(1.13) Remark. Everything in this section (except (1.7) and exercise 7 above) makes sense and is valid for an arbitrary ordered field R, not necessarily real closed.

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§2. Triangulation theorem

First some facts on extending continuous definable functions.

(2.1) LEMMA. *Let σ be a k-simplex in R^m, $k > 0$, and $f : \partial\sigma \to R$ a continuous definable function. Then f has a continuous definable extension $g : \mathrm{cl}(\sigma) \to R$.*

PROOF. Let $b = b(\sigma)$ be the barycenter of σ. Connect each point $p \in \partial\sigma$ to b by the line segment $[p, b] = \{tp + (1-t)b :\ 0 \le t \le 1\}$, and define g on this line segment by $g\big(tp + (1-t)b\big) = tf(p)$. This gives g the value 0 at the barycenter b, and g is well defined on all of $\mathrm{cl}(\sigma)$ since the line segments cover $\mathrm{cl}(\sigma)$ and line segments corresponding to different p's only intersect at the point b. Moreover g extends f. By viewing $\mathrm{cl}(\sigma)$ as the continuous image of the map

$$(t, p) \mapsto tp + (1-t)b \ :\ [0,1] \times \big(\mathrm{cl}(\sigma) - \sigma\big) \to R^m$$

we can appeal to Chapter 6, (1.13) to conclude that g is continuous. $\square$

(2.2) LEMMA. *Let K be a closed complex in R^m and L a closed subcomplex. Then each continuous definable function $f : |L| \to R$ has a continous definable extension $\tilde{f} : |K| \to R$.*

PROOF. It will suffice to show that if $L \ne K$, then we can find a strictly larger closed subcomplex L' of K and a continuous definable extension $f' : |L'| \to R$. Assuming $L \ne K$, take a simplex $\sigma \in K - L$ of minimal dimension. If σ is a 0-simplex, then $\sigma = \{a\}$ for a point $a \notin |L|$, $L \cup \{\sigma\}$ is a strictly larger closed subcomplex of K, and f has continuous definable extension $f' : |L \cup \{\sigma\}| = |L| \cup \{a\} \to R$ given by $f'(a) = 0$. Next assume σ is a k-simplex with $k > 0$. Then all proper faces of σ are in L, so $\mathrm{cl}(\sigma) - \sigma \subseteq |L|$, and by the previous lemma the function $f|\,\mathrm{cl}(\sigma) - \sigma$ extends continuously to a definable function $g : \mathrm{cl}(\sigma) \to R$. Further $L' := L \cup \{\sigma\}$ is a strictly larger closed subcomplex, and f extends continuously to the function $f' : |L'| = |L| \cup \mathrm{cl}(\sigma) \to R$ defined by

$$\begin{aligned} f'(x) &= f(x) \text{ for } x \in |L|, \\ f'(x) &= g(x) \text{ for } x \in \mathrm{cl}(\sigma). \quad \square \end{aligned}$$

(2.3) DEFINITION. Let $A \subseteq R^m$ be a definable set. A **triangulation in R^n** of A is a pair (Φ, K) consisting of a complex K in R^n and a definable homeomorphism $\Phi : A \to |K|$. Note that then

$$\Phi^{-1}(K) := \{\Phi^{-1}(\sigma) :\ \sigma \in K\}$$

is a finite partition of A. We call $\big(A, \Phi^{-1}(K)\big)$ a **triangulated set**. The triangulation is said to be **compatible with the subset** $A' \subseteq A$ if A' is a union of elements of $\Phi^{-1}(K)$. Note that by Chapter 6. (1.10), we have

$$A \text{ is closed and bounded} \Leftrightarrow \text{ the complex } K \text{ is closed.}$$

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In that case each continuous definable R-valued function on $\mathrm{cl}(C)$, where $C \in \Phi^{-1}(K)$, has a continuous definable R-valued extension to A, by lemma (2.2).

(2.4) Let $(A, \mathcal{P})$ be a triangulated set, and $C, D \in \mathcal{P}$.

We call D a **face** of C if $D \subseteq \mathrm{cl}(C)$ and a **proper face** of C if $D \subseteq \mathrm{cl}(C) - C$.

Note that if $\mathcal{P} = \Phi^{-1}(K)$ for the triangulation (Φ, K) of A, then D is a (proper) face of C iff the simplex $\Phi(D)$ is a (proper) face of the simplex $\Phi(C)$. Hence,

If D is a proper face of C and $y \in D$, then there are arbitrarily small definable neighborhoods U of y in A such that $U \cap C$ is definably connected. (To see this, use the fact that a convex definable set is definably connected.)

Note also that $\mathrm{cl}(C) \cap A$ = union of the faces of C in $\mathcal{P}$, for $C \in \mathcal{P}$.

(2.5) Definition. A **multivalued function F on the triangulated set** $(A, \mathcal{P})$ is a finite collection of functions, $F = \{f_{C,i} : C \in \mathcal{P},\ 1 \le i \le k(C)\}$, $k(C) \ge 0$, each $f_{C,i} : C \to R$ continuous and definable, and $f_{C,1} < \cdots < f_{C,k(C)}$. We set

$$\begin{aligned} \Gamma(F) &:= \bigcup_{f \in F} \Gamma(f), \\ F|C &:= \{f_{C,i} : 1 \le i \le k(C)\} \text{ for } C \in \mathcal{P}, \\ \mathcal{P}^F &:= \{\Gamma(f) : f \in F\} \cup \{(f_{C,i}, f_{C,i+1}) : C \in \mathcal{P},\ 1 \le i < k(C)\}, \\ A^F &:= \text{the union of the sets in } \mathcal{P}^F, \end{aligned}$$

so $\mathcal{P}^F$ is a partition of $A^F \subseteq R^{m+1}$ and $\pi(\Gamma(F)) = \pi(A^F) \subseteq A$. Here $\pi : R^{m+1} \to R^m$ is the projection map onto the first m coordinates.

We consider two properties of a multivalued function F on $(A, \mathcal{P})$:

(1) We call F **closed** if for each pair $C, D \in \mathcal{P}$ with D a proper face of C and each $f \in F|C$ there is $g \in F|D$ such that $g(y) = \lim_{x \to y} f(x)$ for all $y \in D$.

Note that then each $f \in F$, say $f \in F|C$, extends continuously to a definable function $\mathrm{cl}(f) : \mathrm{cl}(C) \cap A \to R$, that the restrictions of $\mathrm{cl}(f)$ to the faces of C in $\mathcal{P}$ belong to F, and that $\Gamma(F)$ is closed in $A \times R$. (Lemma (2.6) below is a converse to this.)

(2) We call F **full** if F is closed, $k(C) \ge 1$ for all $C \in \mathcal{P}$, and for each pair $C, D \in \mathcal{P}$ with D a proper face of C and each $g \in F|D$ we have $g = \mathrm{cl}(f)|D$ for some $f \in F|C$, where $\mathrm{cl}(f)$ is the continuous extension of f to $\mathrm{cl}(C) \cap A$.

Note that then $\pi\Gamma(F) = A$, and that if A is closed and bounded, so is A^F.

(2.6) Lemma. *Let F be a multivalued function on the triangulated set $(A, \mathcal{P})$ such that $\Gamma(F)$ is closed in $A \times R$, and there is $M > 0$ such that $\Gamma(F) \subseteq A \times [-M, M]$. Then F is closed.*

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Proof. Let $C, D \in \mathcal{P}$, D a proper face of C, and let $f \in F|C$. Given $y \in D$ we have to show first that $\lim_{x \to y} f(x)$ exists. Set $l := \liminf_{x \to y} f(x)$, $L := \limsup_{x \to y} f(x)$; i.e. $l = \inf\{t \in R : \text{there are } x \in C \text{ arbitrarily close to } y \text{ with } f(x) < t\}$, and similarly for L. Clearly $-M \leq l \leq L \leq +M$, and $l = L$ implies that $\lim_{x \to y} f(x)$ exists and equals l. Suppose $l < L$; take any t with $l < t < L$. By the *remark* in (2.4) there are arbitrarily small definable neighborhoods U of y in A such that $U \cap C$ is definably connected. Since f takes both values $< t$ and values $> t$ on $U \cap C$ it takes also the value t on $U \cap C$. Hence $(y, t) \in \mathrm{cl}(\Gamma(f)) \subseteq \Gamma(F)$ for each t in the interval (l, L). But there can only be finitely many t with $(y, t) \in \Gamma(F)$. This contradiction shows $l = L = \lim_{x \to y} f(x)$. Set $\phi(y) = \lim_{x \to y} f(x)$. One checks easily that $\phi : D \to R$ is continuous, and since $\Gamma(F)$ is closed it follows that $\phi \in F$. $\square$

(2.7) More terminology. Let (Φ, K) be a triangulation in R^n of $A \subseteq R^m$, and let $B \subseteq R^{m+1}$ be a definable set, $B \subseteq A \times R$. Then a triangulation (Ψ, L) in R^{n+1} of B is said to be a **lifting** of (Φ, K) if $K = \{\pi(\sigma) : \sigma \in L\}$ and the diagram

$$\begin{array}{ccc} B & \xrightarrow{\Psi} & |L| \\ \downarrow & & \downarrow \\ A & \xrightarrow[\Phi]{} & |K| \end{array}$$

commutes where the vertical maps are restrictions of the projection maps $R^{m+1} \to R^m$ onto the first m coordinates and $R^{n+1} \to R^n$ onto the first n coordinates. (In particular, B projects onto A.)

The triangulation (Φ, K) of $A \subseteq R^m$ gives rise to the triangulation (Φ, K') of A, where K' is as in (1.8). We call (Φ, K') the **barycentric subdivision of** (Φ, K).

Here is the main triangulation lemma.

(2.8) Lemma. *Let (Φ, K) be a triangulation in R^n of the closed and bounded definable set $A \subseteq R^m$ and let F be a full multivalued function on the triangulated set $(A, \Phi^{-1}(K))$. Then (Φ, K) or its barycentric subdivision (Φ, K') can be lifted to a triangulation (Ψ, L) in R^{n+1} of A^F that is compatible with the sets in $\Phi^{-1}(K)^F$.*

Proof. Note first that K is closed. Replacing K if necessary by its barycentric subdivision K' and F by the corresponding set of restrictions, we may assume in addition to fullness of F that the following condition holds for each $C \in \Phi^{-1}(K)$:

$$(*) \qquad \begin{cases} \text{If } f_1, f_2 \in F|C,\ f_1 \neq f_2, \text{ then there is at least one vertex of} \\ C \text{ where } \mathrm{cl}(f_1) \text{ and } \mathrm{cl}(f_2) \text{ take different values.} \end{cases}$$

Here the vertices of C are $\Phi^{-1}(a_0), \ldots, \Phi^{-1}(a_k)$ with $a_0, \ldots, a_k$ the vertices of $\Phi(C)$.

Fix a linear order on $\mathrm{Vert}(K)$. We now construct L and Ψ above each $C \in \Phi^{-1}(K)$. Let $a_0, \ldots, a_k$ be the vertices of $\Phi(C)$ listed in the order we imposed on $\mathrm{Vert}(K)$.

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Let $f < g$ be two *successive* members of $F|C$. Put

$$r_i := \mathrm{cl}(f)\big(\Phi^{-1}(a_i)\big),\ s_i := \mathrm{cl}(g)\big(\Phi^{-1}(a_i)\big),\ b_i := (a_i, r_i),\ c_i := (a_i, s_i) \in R^{n+1}.$$

Because of $(*)$ the conditions for lemma (1.10) are satisfied. Let $L(f,g)$ be the complex in R^{n+1} constructed in that lemma, so $|L(f,g)|$ is the convex hull of $\{b_0, \ldots, b_k, c_0, \ldots, c_k\}$. Define the map $\Psi_{f,g} : [\mathrm{cl}(f), \mathrm{cl}(g)] \to |L(f,g)|$ by

$$\Psi_{f,g}\big(x, t\,\mathrm{cl}(f)(x) + (1-t)\,\mathrm{cl}(g)(x)\big) \;=\; t\Phi_b(x) + (1-t)\Phi_c(x),\ 0 \le t \le 1,$$

where $\Phi_b(x)$, $\Phi_c(x)$ are the points of $[b_0, \ldots, b_k]$ and $[c_0, \ldots, c_k]$ with the same affine coordinates with respect to $b_0, \ldots, b_k$ and $c_0, \ldots, c_k$ as $\Phi(x) \in [a_0, \ldots, a_k]$ has with respect to $a_0, \ldots, a_k$. Clearly $\Psi_{f,g}$ is a bijection, and it follows from Chapter 6, (1.13)(ii) that $\Psi_{f,g}$ is continuous.

We also define for each $f \in F|C$ the closed complex $L(f)$ in R^{n+1}: it consists of the k-simplex $(b_0, \ldots, b_k)$ with $b_i = \big(a_i, \mathrm{cl}(f)\big(\Phi^{-1}(a_i)\big)\big)$ as above, and all its faces. Then $\Psi_f : \Gamma\big(\mathrm{cl}(f)\big) \to |L(f)|$ is by definition the homeomorphism given by

$$\Psi_f\big(x, \mathrm{cl}(f)(x)\big) \;=\; \Phi_b(x)$$

where $\Phi_b(x)$ is defined as before.

Now let L be the union of all complexes $L(f,g)$ and $L(f)$, for all $C \in \Phi^{-1}(K)$. A lengthy but routine argument using fullness of F shows L is a closed complex in R^{n+1}, and that any two among the $\Psi_{f,g}$'s and Ψ_f's coincide on the intersection of their domains. (In the interest of conciseness we leave here the numerous details to the reader.) Hence these maps glue to a continuous definable map $\Psi : A^F \to |L|$. One verifies easily that Ψ is a bijection. Hence, by Chapter 6, (1.12), Ψ is homeomorphism. Using the last remark of (1.11) it is routine to check that the triangulation (Ψ, L) lifts (Φ, K). Finally, note that (Ψ, L) is compatible with the sets in $\Phi^{-1}(K)^F$. $\square$

(2.9) TRIANGULATION THEOREM. *Let $S \subseteq R^m$ be a definable set, with definable subsets $S_1, \ldots, S_k$. Then S has a triangulation in R^m that is compatible with these subsets.*

PROOF. By induction on m, the case $m = 0$ being trivial. Suppose the triangulation theorem holds for a certain value of m. To prove that each definable set $S \subseteq R^{m+1}$ has a triangulation in R^{m+1} compatible with any given definable subsets $S_1, \ldots, S_k$, we first reduce to the case that S is closed and bounded. Let $\mu : R \to (-1,1)$ be the definable homeomorphism $t \mapsto t/\sqrt{1+t^2}$, and S^μ and S_i^μ the images of S and S_i under the homeomorphism $(x_1, \ldots, x_m) \mapsto \big(\mu(x_1), \ldots, \mu(x_m)\big) : R^m \to (-1,1)^m$. It suffices to prove that the closure of S^μ, a closed and bounded set in R^{m+1}, has a triangulation in R^{m+1} that is compatible with its subsets S^μ and $S_1^\mu, \ldots, S_k^\mu$. So, adjusting our notations, we may as well assume from the start that S is already closed and bounded. Under this assumption on S, let

$$T \;:=\; \mathrm{bd}(S) \cup \mathrm{bd}(S_1) \cup \cdots \cup \mathrm{bd}(S_k),$$

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so $\dim(T) < m+1$ by Chapter 4, (1.10). Note also that T is closed and bounded.

By the good directions lemma (4.2) in Chapter 7 the set T has a good direction vector. Choose a linear automorphism of R^{m+1} that maps this good direction vector to the vertical unit vector e_{m+1}. To obtain a triangulation of S in R^{m+1} compatible with $S_1, \ldots, S_k$ we may replace S and the S_i's by their images under this linear automorphism, which replaces T also by its image, so we may assume

$(*)$ the unit vector e_{m+1} is a good direction for the set T.

Under this assumption we shall prove that there is a triangulation of $\pi(S) \subseteq R^m$ in R^m that can be lifted to a triangulation of S compatible with $S_1, \ldots, S_k$.
Set $A := \pi(S) = \pi(T)$, a closed and bounded subset of R^m. By $(*)$ and cell decomposition the set T is the disjoint union of $\Gamma(f)$'s for finitely many definable continuous functions f on cells A_j that form a partition of A. By the inductive hypothesis there is a triangulation (Φ, K) of A in R^m that is compatible with each set A_j. The nonempty restrictions of the f's to the sets of $\Phi^{-1}(K)$ form a multivalued function F on $\big(A, \Phi^{-1}(K)\big)$ such that $\Gamma(F) = T$. Since T is closed, lemma (2.6) implies that F is closed. Unfortunately, F may not be full.

Therefore we modify F as follows to $\tilde{F}$. Each $f \in F$ extends, first continuously to the closure of its domain, and then, by the last remark of (2.3), further to a continuous definable function $\tilde{f}: A \to R$. Change T to $\tilde{T} :=$ union of the graphs $\Gamma(\tilde{f})$ for $f \in F$, and take a new triangulation $(\tilde{\Phi}, \tilde{K})$ of A in R^m that is compatible with all sets in $\Phi^{-1}(K)$, all sets $\pi\big(S \cap \Gamma(\tilde{f})\big)$ and $\pi\big(S_i \cap \Gamma(\tilde{f})\big)$ for $f \in F$, $i \in \{1, \ldots, k\}$, and such that the nonempty restrictions of the $\tilde{f}$'s (for $f \in F$) to the sets of $\mathcal{P} := \tilde{\Phi}^{-1}(\tilde{K})$ form a multivalued function $\tilde{F}$ on $(A, \mathcal{P})$. Clearly $\Gamma(\tilde{F}) = \tilde{T}$. Since $\tilde{T}$ is closed, $\tilde{F}$ is closed by lemma (2.6), and therefore $\tilde{F}$ is full. Hence by the last lemma we can lift $(\tilde{\Phi}, \tilde{K})$ or its barycentric subdivision to a triangulation (Ψ, L_0) of $A^{\tilde{F}}$ that is compatible with the sets of $\mathcal{P}^{\tilde{F}}$. From the compatibility of $(\tilde{\Phi}, \tilde{K})$ with the sets mentioned it follows that if $B \in \mathcal{P}$ and $g \in \tilde{F}|B$ then $\Gamma(g) \subseteq S$, or $\Gamma(g) \cap S = \emptyset$, and similarly for S_i, $i \in \{1, \ldots, k\}$. Also, if $B \in \mathcal{P}$ and g, h are successive members of $\tilde{F}|B$ it follows from Chapter 6, (3.3), that $(g, h) \subseteq S$ or $(g, h) \cap S = \emptyset$, and similarly for each S_i, $1 \le i \le k$. Hence $\mathcal{P}^{\tilde{F}}$ partitions S and each S_i. Let $L := \big\{\sigma \in L_0 : \sigma \subseteq \Psi(S)\big\}$. Then $(\Psi|S, L)$ is clearly a triangulation of S compatible with $S_1, \ldots, S_k$ and it lifts $(\tilde{\Phi}, \tilde{K})$ or its barycentric subdivision. $\square$

(2.10) The following example shows decisively that the use of a linear automorphism (alias a linear change of coordinates) in the proof above cannot be omitted. Something like the choice of a "good direction" seems essential for triangulation.

Example. Let $S := \big\{(x, y, z) \in R^3 : y = xz,\ |x|, |z| \le 1\big\}$. Then, with $\pi: R^3 \to R^2$ the usual projection map we have $\pi(S) = \big\{(x, y) \in R^2 : |y| \le |x| \le 1\big\}$. We claim

No triangulation of $\pi(S)$ can be lifted to a triangulation of S.

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To see this, consider a triangulation (Φ, K) of $\pi(S)$, and suppose this triangulation can be lifted to a triangulation $(\hat{\Phi}, \hat{K})$ of S. We shall derive a contradiction. Let $\mathcal{P} := \Phi^{-1}(K)$ and $\hat{\mathcal{P}} := \hat{\Phi}^{-1}(\hat{K})$, so $\mathcal{P} = \{\pi(A) : A \in \hat{\mathcal{P}}\}$. Let $T := \{(0,0,z) : |z| \leq 1\} \subseteq S$. Then T is a union of sets in $\hat{\mathcal{P}}$ since $(0,0)$ must be a vertex of $\mathcal{P}$ (exercise). In particular, T contains a one-dimensional set $A \in \hat{\mathcal{P}}$, corresponding to a 1-simplex under $\hat{\Phi}$. Let $a, b \in T$ be the two vertices of A.

Since A is not open in S, it is a proper face of a set $B \in \hat{\mathcal{P}}$ corresponding to a 2-simplex under $\hat{\Phi}$ (exercise). Then $\dim(B - T) = 2$, and π is injective on $S - T$, so $\dim(\pi(B)) = 2$. Hence $\pi(B) \in \mathcal{P}$ corresponds to a 2-simplex under Φ, and the vertices of $\pi(B)$ are the images under π of the vertices of B. But π collapses the distinct vertices a, b of B to the single point $(0,0)$ in $\pi(B)$, so $\pi(B)$ cannot have three distinct vertices, contradiction.

(2.11) Digression 1: criterion for definable equivalence.

Call definable sets $A \subseteq R^m$ and $B \subseteq R^n$ **definably equivalent** if there is a definable bijection from A onto B. Clearly "definable equivalence" is an equivalence relation on the class of definable sets.

Corollary. *Let $A \subseteq R^m$ and $B \subseteq R^n$ be definable sets. Then*

$$A \text{ and } B \text{ are definably equivalent } \Leftrightarrow \dim(A) = \dim(B) \text{ and } E(A) = E(B).$$

Proof. The $\Rightarrow$-direction follows from Chapter 4. Suppose now that A and B have the same dimension and the same Euler characteristic. By our triangulation theorem we may also assume that A and B are finite disjoint unions of simplexes in R^m and R^n respectively. For this case we prove a bit more:

Claim. A and B are semilinearly equivalent, that is, there is a semilinear bijection from A onto B.

Before starting the proof of this claim, first three general observations.

(1) A k-simplex is, for $k > 0$, the disjoint union of two k-simplexes and a $(k-1)$-simplex. (Exercise, induction on k.)
(2) Any two k-simplexes are semilinearly homeomorphic. (By remark in (1.2).)
(3) $E(\sigma) = (-1)^k$ for a k-simplex σ. (See exercise 1 below.)

Proof of claim. By induction on $k = \dim(A) = \dim(B)$. For $k = 0$ the sets A and B are finite of the same cardinality $E(A) = E(B)$, and any bijection will do. Let $k > 0$. Write A and B as finite disjoint unions of simplexes:

$$\begin{aligned} A &= \sigma_1 \cup \cdots \cup \sigma_p \cup \cdots \cup \sigma_q, \\ B &= \tau_1 \cup \cdots \cup \tau_r \cup \cdots \cup \tau_s, \end{aligned}$$

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where the σ_i and τ_j are of maximal dimension k for $1 \leq i \leq p$ and $1 \leq j \leq r$, and of lower dimension for $i > p$ and $j > r$.

Using observation (1) we may arrange that $p = r$, and that there is at least one $(k-1)$-simplex among the σ_i for $i > p$, as well as at least one $(k-1)$-simplex among the τ_j for $j > p$. By observation (2) there is a semilinear bijection between $\sigma_1 \cup \dots \cup \sigma_p$ and $\tau_1 \cup \dots \cup \tau_p$. Also $A' := \sigma_{p+1} \cup \dots \cup \sigma_q$ and $B' := \tau_{p+1} \cup \dots \cup \tau_s$ have both dimension $k-1$, and the same Euler characteristic, by observation (3). So by the inductive hypothesis, A' and B' are semilinearly equivalent. Hence A and B are semilinearly equivalent. □

(2.12) DIGRESSION 2: NUMBER OF DEFINABLE HOMEOMORPHISM TYPES.

Let us say that definable sets $A \subseteq R^m$ and $B \subseteq R^n$ are **definably homeomorphic** if there is a definable homeomorphism $f : A \to B$. This defines an equivalence relation on the collection of definable sets in R^m, $m = 0, 1, 2, \dots$, and a **definable homeomorphism type** is simply an equivalence class for this equivalence relation. An easy consequence of the triangulation theorem is that *there are only countably many definable homeomorphism types*. To see this, suppose that a (nonempty) definable set $A \subseteq R^m$ is definably homeomorphic to $|K|$ where K is a complex with N vertices. Let $e_1, \dots, e_N$ be the basis vectors of R^N.

CLAIM. A is definably homeomorphic to a union of faces of the simplex $(e_1, \dots, e_N)$.

Just take a bijection $V : \mathrm{Vert}(K) \to \{e_1, \dots, e_N\}$, and let L be the complex with $\mathrm{Vert}(L) = \{e_1, \dots, e_N\}$ such that $V : K \to L$ is a vertex map whose inverse is also a vertex map; then $|V| : |K| \to |L|$ is a definable homeomorphism, and the claim is an obvious consequence.

(2.13) We close this section with some exercises. Exercise 3 is intended for readers familiar with elementary model theory. Moreover, exercise 4 depends on the result of exercise 3. Actually, in the next chapter we obtain essentially the same results as stated in the exercises without any model theory. I include the exercise nevertheless, since it illustrates a typical model-theoretic method.

(2.14) EXERCISES.

1. Let Δ_m be the m-simplex in R^m spanned by $0, e_1, \dots, e_m$, where $e_1, \dots, e_m$ are the standard basis vectors. Show that

$$\Delta_{m+1} = \left\{ (x, r) \in R^{m+1} : \ x \in \Delta_m, \ 0 < r < 1 - \sum x_i \right\}.$$

Derive from this that Δ_m is an open cell in R^m, and that each m-simplex in R^m is definably homeomorphic to an open cell in R^m.

2. Show that each bounded semilinear set is a polyhedron. More generally, let $S_1, \dots, S_k$ be semilinear subsets of a bounded semilinear set $S \subseteq R^m$. Show that

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there is a complex K in R^m such that $S = |K|$, and each S_i is a union of simplexes of K. (The result of this exercise holds for any ordered field R.)

3. Let $S \subseteq R^{m+n}$ be definable. Show that the sets S_a fall into only finitely many definable homeomorphism types as the parameter a ranges over R^m. More precisely, show there is a definable map $f: S \to R^N$ (for some N) such that for each $a \in R^m$ the map $f_a: S_a \to R^N$ is a homeomorphism from S_a onto a union of faces of $(e_1, \ldots, e_N)$.

4. Let $S \subseteq R^{m+n}$ be definable. Show that there are a partition of R^m into definable sets $A_1, \ldots, A_k$ and for each $i = 1, \ldots, k$ a definable set $F_i \subseteq R^n$ and a definable homeomorphism $h_i : S \cap (A_i \times R^n) \xrightarrow{\sim} A_i \times F_i$ such that the diagram

$$\begin{tikzcd} S \cap (A_i \times R^n) \arrow[rr, "\sim", "h_i"'] \arrow[dr] & & A_i \times F_i \arrow[dl] \\ & A_i & \end{tikzcd}$$

commutes. (Here the downward arrows indicate the obvious projection maps.) Hint: use the result of exercise 3 in combination with Chapter 6, (2.5), exercise 2.

§3. Definable retractions and definable continuous extensions

As one might expect, the triangulation theorem has numerous consequences, and we develop a few here that are needed in later chapters.

(3.1) HOMOTOPIES, RETRACTIONS AND EXTENSIONS.

Let $A \subseteq R^m$ and $B \subseteq R^n$ be definable sets, and $f, g: A \to B$ definable continuous maps. A **definable homotopy between f and g** is a definable continuous map $H: A \times [0,1] \to B$ such that $f(x) = H(x,0)$ and $g(x) = H(x,1)$ for all x in A. (It may help to view H as a "continuous" family of maps $(H_t)_{0 \le t \le 1}$, with $H_t: A \to B$ given by $H_t(x) = H(x,t)$, so $f = H_0$ and $g = H_1$.)

A definable set A is called **definably contractible to the point** $a \in A$ if there is a definable homotopy $H: A \times [0,1] \to A$ between the identity map on A and the map $A \to A$ taking the constant value a. (Such an H is called a **definable contraction from A to the point** a.)

Note that for a point $a \in R^m$ the map $H: R^m \times [0,1] \to R^m$ given by

$$H(x,t) = (1-t)x + ta$$

is a definable contraction from R^m to a. Since each cell is definably homeomorphic to some R^m it follows that each cell is definably contractible to each of its points.

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Let a definable subset A' of a definable set $A \subseteq R^m$ be given, and denote by $i_{A'}$ the inclusion map $A' \to A$.

A map $r : A \to A'$ is called a **definable retraction** if r is a definable continuous map such that $r(x) = x$ for all $x \in A'$. So a definable retraction $r : A \to A'$ satisfies $r \circ i_{A'} = 1_{A'}$.

Next we define a **definable contraction from** A **to** A' to be a definable homotopy $H : A \times [0,1] \to A$ between 1_A and $i_{A'} \circ r$, for a definable retraction $r : A \to A'$. This map r is then uniquely determined by $r(x) = H(x,1)$, and if we want to be more explicit, we call H a **definable contraction from** A **to** r. If in addition $H(a',t) = a'$ for all $a' \in A'$ and $t \in [0,1]$ we call H a **definable strong deformation retraction from** A **to** A'. Note that for $a \in A$ a definable contraction from A to the point a is the same as a definable contraction from A to the subset $\{a\}$.

We also need one important construction in complexes:

(3.2) Let K be a complex.

Given a definable set $A \subseteq |K|$ we define the **star** $st_K(A)$ of A in K to be the union of all simplexes $\sigma \in K$ such that $\mathrm{cl}(\sigma) \cap A \neq \emptyset$. Clearly $st_K(A)$ is an open neighborhood of A in $|K|$, and in fact the smallest neighborhood of A in $|K|$ that is a union of simplexes of K.

We can now formulate a basic result on the existence of nice retractions.

(3.3) PROPOSITION. *Let K be a complex and L a subcomplex of K, closed in K. Then there is a definable retraction $r : st_{K'}(|L|) \to |L|$ such that for each point $x \in st_{K'}(|L|) - |L|$ the open line interval $(x, r(x))$ lies entirely in the simplex of K' that contains the point x.*

(3.4) REMARKS.

(1) We shall give an explicit definition of the retraction r.

(2) Given such r the map $H : st_{K'}(|L|) \times [0,1] \to st_{K'}(|L|)$ given by

$$H(x,t) = (1-t)\cdot x + t \cdot r(x)$$

is a definable strong deformation retraction from $st_{K'}(|L|)$ to $|L|$. The image $r(S)$ of each simplex $S \subseteq st_{K'}(|L|)$ of K' is contained in $\mathrm{cl}(S)$.

(3.5) Before proving (3.3) we need some preparations. Let $E := \mathrm{Vert}(K) = \mathrm{Vert}(\overline{K})$. For each vertex $e \in E$ we define a function $\lambda_e : |\overline{K}| \to [0,1]$ by setting for a point $x \in (e_0, \dots, e_k)$, for a k-simplex $(e_0, \dots, e_k) \in \overline{K}$,

$$\begin{aligned} \lambda_e(x) \ &:= \ 0 \text{ if } e \notin \{e_0, \dots, e_k\}, \\ &:= \ \lambda_i \text{ if } e = e_i, \text{ where } x = \sum \lambda_j e_j,\ \sum \lambda_j = 1, \\ &\qquad \text{with } 0 < \lambda_j \text{ for } j \in \{0, \dots, k\}. \end{aligned}$$

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We call λ_e the **affine coordinate function of $\overline{K}$ corresponding to the vertex** e. Note that λ_e is definable and continuous, and for each $x \in |\overline{K}|$ we have

$$x = \sum_{e \in E} \lambda_e(x) \cdot e,$$
$$1 = \sum_{e \in E} \lambda_e(x).$$

Now $\{b(\sigma) : \sigma \in \overline{K}\}$ is the set of vertices of the first barycentric subdivision $\overline{K}'$ of $\overline{K}$, and so we also have for each $\sigma \in \overline{K}$ the affine coordinate function $\lambda_{b(\sigma)}$ of $\overline{K}'$ corresponding to the vertex $b(\sigma)$. For convenience we denote $\lambda_{b(\sigma)}$ by λ_σ, so $\lambda_\sigma : |\overline{K}'| = |\overline{K}| \to [0,1]$ is a definable (continuous) function.

Next we introduce for $\sigma \in \overline{K}$ a "weight function" $w_\sigma : |\overline{K}| \to [0,1]$ as follows:

(i) if $\sigma \in L$, then w_σ is identically 1,
(ii) if $\sigma \in \overline{K} - \overline{L}$, then w_σ is identically 0,
(iii) if $\sigma \in \overline{L} - L$, then

$$w_\sigma(x) = \frac{\sum_{\sigma < \tau \in L} \lambda_\tau(x)}{\sum_{\sigma < \tau \in K} \lambda_\tau(x)},$$

provided the denominator $\sum_{\sigma < \tau \in K} \lambda_\tau(x)$ is nonzero, and $w_\sigma(x) = 0$ otherwise.

(The summations are over the $\tau \in L$ with $\sigma < \tau$, respectively over the $\tau \in K$ with $\sigma < \tau$.) Clearly w_σ is definable, though it may not be continuous.

Now define $\mu_\sigma : |K| \to [0,1]$ by $\mu_\sigma(x) := w_\sigma(x)\lambda_\sigma(x)$. Clearly μ_σ is definable, and by the following lemma μ_σ is also continuous.

(3.6) LEMMA. *Let $\sigma \in \overline{L} - L$, and let $S \in K'$ be a simplex on which the function $\sum_{\sigma < \tau \in K} \lambda_\tau$ vanishes. Then λ_σ also vanishes on S.*

PROOF. Let $S = (b(\sigma_0), \ldots, b(\sigma_n))$ with $\sigma_0 < \cdots < \sigma_n$ a K-flag, $\sigma_n \in K$. Suppose λ_σ is nonzero at some point of S, hence at every point of S. Then σ is among $\sigma_0, \ldots, \sigma_{n-1}, \sigma_n$, but $\sigma = \sigma_n$ would imply $\sigma \in K \cap \overline{L} = L$, contradiction. Hence σ is among $\sigma_0, \ldots, \sigma_{n-1}$, in particular, $n \geq 1$. Thus $\sigma < \sigma_n \in K$, so that

$$\sum_{\sigma < \tau \in K} \lambda_\tau(x) > 0,$$

for all $x \in S$. $\square$

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(3.7) LEMMA. *For each $x \in st_{K'}(|L|)$ there is $\sigma \in L$ with $\mu_\sigma(x) > 0$.*

PROOF. Let $S \in K'$ be contained in $st_{K'}(|L|)$. As above, $S = (b(\sigma_0), \ldots, b(\sigma_n))$ for some K-flag $\sigma_0 < \cdots < \sigma_n$, $\sigma_n \in K$. Some face of S intersects $|L|$, hence $\sigma_k \in L$ for some $k \in \{0, \ldots, n\}$. Then we have for $\sigma = \sigma_k$: $\mu_\sigma(x) = \lambda_\sigma(x) > 0$ for all $x \in S$. $\square$

We continue the proof of proposition (3.3). Suppose the complex K lies in R^m. Then we define the (definable, continuous) map $r : st_{K'}(|L|) \to R^m$ by

$$(*) \qquad r(x) = \frac{\sum \mu_\sigma(x) \cdot b(\sigma)}{\sum \mu_\sigma(x)}, \text{ with } \sigma \text{ ranging over } \overline{K}.$$

(3.8) LEMMA. *$r(S) \subseteq \mathrm{cl}(S) \cap |L|$ for each $S \in K'$ contained in $st_{K'}(|L|)$, and $r(x) = x$ for $x \in |L|$.*

Thus $r(st_{K'}(|L|)) \subseteq |L|$ and for $x \in st_{K'}(|L|) - |L|$ the open line interval $(x, r(x))$ lies entirely in the simplex $S \in K'$ that contains the point x. So this lemma implies that r is a retraction with the properties stated in proposition (3.3).

PROOF OF THE LEMMA. As in the previous lemma, let $S = (b(\sigma_0), \ldots, b(\sigma_n))$ with $\sigma_0 < \cdots < \sigma_n$, $\sigma_n \in K$. Take $k \in \{0, \ldots, n\}$ maximal with $\sigma_k \in L$. Consider a $\sigma \in \overline{K}$ such that μ_σ does not vanish identically on S. Then λ_σ does not vanish on S, so $\sigma = \sigma_t$ for some t. If $t > k$ the function w_σ, and hence also μ_σ would vanish identically on S. Thus $t \leq k$. This gives for each $x \in S$ a convex combination $r(x) = \sum_{0 \leq i \leq k} \alpha_i b(\sigma_i)$ with $0 < \alpha_i \in R$ for all i and $\sum \alpha_i = 1$, as is easily checked. Hence $r(S) \subseteq (b(\sigma_0), \ldots, b(\sigma_k)) \subseteq \mathrm{cl}(S) \cap |L|$. If $S \subseteq |L|$, then $\sigma_n \in L$, so $k = n$, and by distinguishing the cases $\sigma \in L$, $\sigma \in \overline{K} - \overline{L}$, and $\sigma \in \overline{L} - L$, we see that in each case $\mu_\sigma | S = \lambda_\sigma | S$. (For the case $\sigma \in \overline{L} - L$, use the fact that $K \cap \overline{L} = L$.) Thus $r(x) = x$ for all $x \in |L|$. $\square$

This finishes the proof of proposition (3.3). In combination with triangulability it gives the following.

(3.9) COROLLARY. *Let A be a definable closed subset of the definable set $B \subseteq R^m$. Then there are a definable open subset U of B containing A, and a definable retraction $r : \mathrm{cl}(U) \cap B \to A$.*

PROOF. By (3.3) there are a definable open neighborhood V of A in B and a definable retraction $r_V : V \to A$. Applying Chapter 6, (3.5) to the closed subsets A and $B - V$ of B we find definable U, open in B and containing A, such that $\mathrm{cl}(U) \cap B \subseteq V$. Now restrict r_V to $\mathrm{cl}(U) \cap B$. $\square$

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Here is an attractive further consequence, generalizing (2.2). It will be needed as a technical tool in the next chapter, on trivialization, to make a multivalued function full.

(3.10) COROLLARY. *Let A be a definable closed subset of the definable set $B \subseteq R^m$. Then each definable continuous function $f: A \to R$ can be extended to a definable continuous function $\tilde{f}: B \to R$. More generally, let $f: A \to C$ be a definable continuous map into a definable set $C \subseteq R^n$ that is definably contractible to a point $c \in C$. Then f can be extended to a definable continuous function $\tilde{f}: B \to C$.*

PROOF. Choose U and r as in (3.9). By Chapter 6, (3.8) there is a definable continuous function $\lambda: B \to [0,1]$ with $\lambda^{-1}(0) = A$ and $\lambda^{-1}(1) = B - U$. Finally, let Φ be a definable contraction from C to c. Then we define the extension $\tilde{f}: B \to C$ of f as follows:

$$\begin{aligned} \tilde{f}(x) &:= \Phi\big(f\big(r(x)\big), \lambda(x)\big), \text{ for } x \in \mathrm{cl}(U) \cap B, \\ &:= c, \text{ for } x \in B - U. \end{aligned}$$

One verifies easily that $\tilde{f}$ has the desired properties. □

Notes and comments

Section 2 of this chapter follows closely the proof of the semialgebraic triangulation theorem by Delfs and Knebusch in [11]. The main difference is that the good directions lemma is only a *partial* substitute for the use of Noether normalization in their proof. The way out of this difficulty is to first establish lemma (2.2) and use that to construct a full multivalued function as starting point for the inductive triangulation procedure. The idea of using multivalued functions and lemma (2.6) was suggested to me by reading some (unpublished?) lecture notes on subanalytic sets by H. Sussmann (mid 1980s).

The claim in (2.11) that two bounded semilinear sets are semilinearly equivalent if and only if they have the same dimension and the same Euler characteristic was made to me in conversation by Schanuel; see [51] for related results.

Section 3 of this chapter is adapted from the semialgebraic case treated by Delfs and Knebusch in [13], which contains further interesting material that extends easily to our setting.

See Dieudonné [16] for some history of triangulation results for manifolds and algebraic varieties, and Giesecke [27], Lojasiewicz [39] and Hironaka [31] for various kinds of triangulation theorems for semialgebraic and semianalytic sets.

The inductive way triangulation was established in this chapter provides extra information that has been used by Woerheide in [65] to prove the excision property for simplicial and singular homology for definable sets in the o-minimal setting. (This

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way of obtaining the excision property seems to be new also in the semialgebraic case. This case is extensively discussed by Knebusch in [34].)

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